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Supersedes	Product Brief Version 1.0 (April 2026)
Version	2.0 — April 2026
Classification	Confidential
Market	Abuja, Nigeria (initial rollout)
Platform	Web application (PWA for Inspector Portal)

Version 2.0 — Change Log

This version incorporates all product design revisions agreed following the v1.0 review. Changes affect the fee structure, verification model, notification channels, onboarding flow, AI architecture, result ranking, valuation report scope, conversation memory model, transaction intermediation, and the addition of short-let in-booking messaging.

Area	Change
Listing fee	Removed entirely. All listings free to upload. BeeBop earns only at transaction completion.
Rent fee structure	Two tiers: up to 5m Naira annual rent — 20k flat (landlord 15k, seeker 5k) + 3% annual rent. Above 5m — 20k flat + 2% annual rent. No cap.
Student accommodation fee	2.5% from accommodation owner + 3% from seeker. Charged independently per party at agreement signing.
Short-let fee	Host always pays 3%. Seeker pays 10% on bookings up to 150,000 Naira total, 8% on bookings above 150,000 Naira.
Sales fee	3% of sale value up to 50m Naira, 2.5% above. Paid entirely by seller. Buyer has zero platform fees. Paystack invoice at signing; agreement withheld until payment confirmed.
Valuation report	Fair market value indicator and comparable listings removed. Report contains only: area infrastructure scores, inspector professional note, and report date.
SMS notifications and verification	Removed. Replaced with email (Resend) for OTP and formal documents, and WhatsApp Business API for time-sensitive notifications.
Seeker onboarding	Abuja area dropdown removed from Step 4. Category preference is the only question. Email is required (used for OTP and document delivery).
Conversation memory	Session-only memory. No persistent chat history across sessions. No Chats sidebar section. Session context in Redis expires on session end or 30 minutes of inactivity.
Result ranking	Saves and enquiries removed as ranking signals. Rating score weighted as second signal after verification status.
Short-let messaging	In-booking group chat added: seeker, host, and BeeBop moderator. Content filtering flags off-platform attempts. On-platform action shortcuts embedded in chat.
Sales commission payer	Confirmed: seller pays 100% of sales commission. Buyer has zero platform fees.

1. Executive Summary

BeeBop is a web-based conversational property marketplace designed specifically for the Nigerian real estate market, with initial operations in Abuja. It blends the natural-language interface of a large-language-model chat tool with the familiar browsing experience of a conventional property marketplace, covering four property categories: off-campus student accommodation, short-let, long-term rent, and property sales.

The platform is built on two foundational principles that no existing Nigerian listing site has operationalised at scale: trust through professional verification, and full intermediation of every transaction. BeeBop does not facilitate direct communication between seekers and landlords for rent or sale properties. Every enquiry, visit, offer, and agreement flows through the platform, ensuring all transaction value remains on BeeBop.

Trust is delivered through a dual-badge verification system backed by a licensed real estate surveying and valuation firm. Properties earn a document verification badge through authenticated title document review and a physical inspection badge through a professional on-site assessment that also captures area-level infrastructure data — building towards the most comprehensive knowledge base of Abuja's residential infrastructure in existence.

Strategic Position

BeeBop enters a market with no established conversational property platform and widespread distrust of existing listing sites. The combination of professional verification, complete transaction intermediation, and AI-powered natural language discovery creates a defensible competitive moat that competitors cannot replicate quickly.

2. Product Architecture

The platform is structured across five layers, each with distinct responsibilities that can scale independently.

Layer	Components	Primary Function
Users	Seekers, Landlords and Agents, Inspectors, Admins, Trusted Agents	Role-based access and permissions
Interface	Conversational UI, Category Browse (Grid), Manual Filters	Discovery and search experience
Core Platform	AI/NLP Engine, Listings, Search and Matching, Verification	Business logic and data processing
Backend Services	Auth and Identity, Notifications, Payments, Analytics	Infrastructure and operations
Data Layer	Listings DB, Users DB, Media Store, Analytics and Knowledge Base	Persistence and infrastructure intelligence

2.1 User Roles

Role	Access	Key Capabilities
Seeker	Public platform	Conversational search, category browse, save listings, send offers, book visits (rent/sale), sign agreements, view dashboard.
Landlord / Agent	Public platform (lister view)	Create free listings (all types), submit documents, manage inventory, respond to offers, sign agreements, manage dashboard.
Inspector	Internal staff portal (restricted)	Receive property assignments, conduct site visits, submit property assessment reports and area infrastructure scores. Cannot commit to database — all submissions routed to admin.
Trusted Agent	Internal staff portal (restricted)	Receive visit assignments, access briefing packs, attend property visits as BeeBop representative, file post-visit reports.
Admin	Internal staff portal (full access)	All inspection and agent capabilities plus: document review queue, inspection report review, badge issuance, visit assignment (manual), listing management, account management, moderation.

3. Verification System

Verification is BeeBop's primary trust mechanism and its most significant competitive differentiator. Two independent badge tracks exist. A listing may earn one, both, or neither. The document badge is a prerequisite for the physical inspection badge. Student accommodation listings are exempt from the document badge requirement.

3.1 Document Verification Badge

Step	Action	Party
1	Landlord submits title documents on listing creation: Certificate of Occupancy, deed of assignment, survey plan, or equivalent appropriate to property type.	Landlord
2	Documents enter the admin review queue in the internal portal.	System
3	Admin manually reviews documents for authenticity — confirming legitimacy, not legal validity — and cross-references against listing details.	Admin
4	If documents are queried, landlord notified via email and WhatsApp to respond or resubmit.	Admin / Landlord
5	On approval, document verification badge issued. Listing status updated. Landlord notified via email and WhatsApp.	Admin

Student Accommodation Exception

Off-campus student accommodation listings are exempt from the document badge requirement. Verification for this category is achieved entirely through the physical inspection process, reflecting the practical reality that student housing operators typically do not hold equivalent title documentation to residential landlords.

3.2 Physical Inspection Badge

The physical badge is issued following a professional site visit by a qualified inspector from BeeBop's affiliated real estate surveying and valuation firm. The visit combines a property assessment — which determines badge eligibility — with a separate area infrastructure assessment published as living data alongside the listing.

Property Assessment — Badge-Linked

Category	What is Assessed
Identity and existence	Physical address matches listing. Property boundaries confirmed on the ground.
Listing accuracy	Photos match the actual property. Room count, size, and layout are as described.
Amenities confirmation	Each listed amenity physically confirmed as present, not confirmed, or absent. Generator, borehole, parking, security — each item assessed individually.
Structural condition	Professional assessment of walls, roof, fittings, and finishes, leveraging the inspector's surveying qualifications.

Area Infrastructure Scoring — Separate Living Layer

Infrastructure scores are published as a continuously updated data layer attached to listings. They never affect badge status. Admin can update scores at any time without requiring a new inspection — for example, following a road rehabilitation or EKEDC supply change. Each category is scored 1 to 5 with a visible timestamp.

Category	What is Measured
Road condition	Surface quality, flood risk, accessibility, proximity to major roads.
NEPA / PHCN electricity supply	Estimated daily supply hours observed by inspector plus landlord self-reported figure. Both values recorded where they differ.
Security and neighbourhood	Observed neighbourhood character, security presence, noise environment.
Proximity and access	Distance to key landmarks: markets, hospitals, universities, major roads — recorded as factual data.

Long-Term Vision — The BeeBop Knowledge Base

Each inspection adds a structured record to BeeBop's growing knowledge base. Over time, repeated inspections build a longitudinal dataset tracking infrastructure change across Abuja's residential areas. This dataset has significant standalone value as a published research and intelligence product — positioning BeeBop as the authoritative source on Abuja's living infrastructure.

3.3 Badge Validity and Revocation

Badge	Validity	Immediate Revocation Triggers	Staged Review Triggers
Document badge	24 months. Landlord resubmits documents at renewal.	Confirmed fraudulent documents. Court order or property injunction filed. Relisting a previously suspended property under a different account.	Three verified seeker complaints within 12 months triggers automatic suspension pending investigation.
Physical badge	12 months. Annual re-inspection required. Inspections batched by area for efficiency.	Confirmed landlord fraud. Significant deliberate misrepresentation discovered post-badge.	Sustained pattern of verified complaints. Significant discrepancy found between listing and inspector findings post-issuance.

Infrastructure scores are never grounds for badge revocation. A property experiencing a local grid failure does not lose its physical badge — the area score is updated, keeping badge and living data cleanly separated.

4. Internal Staff Portal

The internal staff portal is a restricted section of the BeeBop website. It is not linked from any public-facing page and is inaccessible without a staff account. Staff accounts are created exclusively by admins — there is no self-registration path. Two roles access the portal: Inspector and Admin.

4.1 Role Permissions

Capability	Inspector	Admin
View own assigned listings only	Yes	N/A — sees all
Mark property GPS location on map	Yes	Yes
Upload photos and video with automatic metadata (GPS, timestamp)	Yes	Yes
Fill property assessment form	Yes	Yes
Fill area infrastructure scoring form (independent of assessment)	Yes	Yes
Submit report to admin review queue	Yes	Yes
View status of own past submissions	Yes	Yes
View all submitted reports	No	Yes
Approve, query, or reject inspection reports	No	Yes
Commit badge to live listing	No	Yes
Publish and update area infrastructure scores independently	No	Yes
Manage all listings	No	Yes
Manage user and landlord accounts	No	Yes
Create and manage inspector and agent accounts	No	Yes
Manually assign visits to trusted agents	No	Yes
Access platform analytics	No	Yes
Moderate short-let in-booking group chats	No	Yes

4.2 Inspector Field Tool — Offline Capability

The inspector portal is built as a Progressive Web Application (PWA) to support field use in areas with unreliable connectivity. The inspection form, photo capture, and GPS tagging functions operate fully offline. All data entered offline is stored locally on the device via IndexedDB and synchronised automatically when connectivity is restored. A visible sync status indicator ensures inspectors always know when their submission has been successfully uploaded.

4.3 Admin Approval Gate

No inspector or agent submission reaches the live database or affects any public-facing listing without explicit admin approval. The admin review queue is the mandatory gate for all field submissions. This protects the integrity of the knowledge base and ensures that every badge issued and every area score published has been reviewed by a human before it is visible to seekers.

5. User Journeys

5.1 Seeker Onboarding

Designed to reach the first value moment — a personalised listing feed — in under two minutes. Budget preference is not collected at onboarding, allowing seekers to browse and dream without self-limiting constraints.

Step	Screen	Detail
1	Email entry	Email required. Used for OTP verification and formal document delivery throughout the platform lifecycle.
2	OTP verification	Six-digit code via Resend email. Resend timer with re-request option.
3	Basic identity	First name and last name. No other fields required at this stage.
4	Category preference	Single question: What are you looking for? Multi-select: Off-campus, Short-let, Rent, For Sale. No area dropdown. No budget field.

Student Seeker Variant

Steps 1 to 3 identical. Step 4 collects: enrolled institution (searchable from BeeBop's institution list), academic level, and gender. Gender is applied silently to room-level filtering throughout the platform — female seekers never see male-tagged rooms and vice versa. The student never sees or manages this filter directly.

5.2 Landlord Onboarding

Step	Screen	Detail
1	Email entry	Email required as primary contact and OTP channel.
2	OTP verification	Via Resend email. WhatsApp OTP available as alternative.
3	Account type	Individual landlord or agency. Determines the verification path.
4a (individual)	Identity verification	Full name, date of birth, NIN verified in real time via NIMC API. BVN collected optionally as secondary layer.

4b (agency)	Business verification	Business name, CAC registration number verified via CAC API. Primary contact NIN.
5	Profile setup	Profile photo (optional but encouraged). Bio (optional). Operating area (optional). For agencies: logo, display name, size, areas of operation.
6	First listing prompt	Immediate prompt to list the first property: List now (flows to listing creation) or Later. Converting at this moment significantly improves completion rates.

5.3 Seeker Discovery — Two Modes

Mode 1 — Conversational Search (Main Page)

The main page presents a centred AI chat input labelled Ask BeeBop. Suggestion pills below scope the query by category: Off-campus, Short-let, Rent, For Sale. A featured listings carousel of verified properties provides immediate value without requiring a query. Unauthenticated visitors see the interface but are prompted to register when they attempt to send a query. The results window has three states: collapsed (first few results), expanded (full grid), minimised (chat input only). There is no persistent chat history — each session is independent.

Mode 2 — Category Browse (Sidebar Navigation)

Selecting a category from the sidebar takes the seeker to a dedicated browse page with a left filter panel and listing grid. A map toggle switches to a pin view of Abuja colour-coded by verification status. The two modes are interconnected: a conversational query can seed a category browse, and a category page contains a prompt to Ask BeeBop to refine results.

5.4 Rent and Sale Transaction Journey

Stage	Actor	Action
View listing	Seeker	Opens listing page. Reviews gallery, description, amenities, badges, area scores, valuation report (if logged in), map, and reviews.
Choose action	Seeker	From the fixed bottom CTA bar: Send an offer, or Schedule a visit.

Offer submitted	Seeker	Offer submitted with price, move-in date, and optional conditions. Sent to landlord via BeeBop.
Landlord responds	Landlord	Accepts, rejects, or sends counter-offer. 48-hour response window. Maximum 3 negotiation rounds.
Offer accepted — visit triggered	System	Accepted offer triggers a visit request in the admin visit queue if no visit has been conducted by this seeker on this listing.
Agent assigned	Admin	Admin manually selects and assigns a trusted agent from the internal portal visit queue.
Visit coordination	Agent	Agent contacts landlord by phone to fix date and time. Seeker notified of confirmed details.
Visit conducted	Agent + Seeker	Agent attends. Landlord preferably absent. Seeker never meets landlord directly.
Post-visit offer	Seeker	Seeker must submit an offer through BeeBop following any visit. Offer button unlocked only after admin approves the post-visit report.
Agreement generated	System	Pre-filled tenancy or sale agreement generated from verified platform data.
Agreement signed	Both parties	Both parties sign via OTP on registered email or WhatsApp. Timestamped and stored.
Fees collected	System	Facilitation fee charged at point of signing via Paystack.
Deal closed	System	Listing marked Let agreed or Sale agreed. Both parties receive agreement copy by email.

5.5 Short-Let Booking Journey

Stage	Detail
Date selection	Seeker selects check-in and check-out from the availability calendar on the listing page. Unavailable dates grayed out. Price updates dynamically. Minimum stay enforced.
Instant booking or request	If host has enabled instant booking, confirmation is immediate on payment. Otherwise, a booking request is sent to the host for acceptance.
Payment captured	Paystack payment at time of confirmation. BeeBop holds funds. Seeker fee (10% or 8% depending on booking value) added to total at checkout.
Check-in	Access details released to seeker on check-in day via email and WhatsApp.
Payout	Host payout after check-in confirmation. Gross booking value minus 3% host fee.

Short-Let — No Visit Option

Short-let listings do not offer a Schedule a Visit option. The only CTA is Reserve. This is a deliberate product decision — the short-let booking flow runs like an online hotel reservation, not a property transaction.

5.6 Student Accommodation Booking

- Institution selector in the CTA bar pre-filtered to institutions that specific accommodation accepts. A BAZE student cannot enquire about a UNIABUJA-only hostel.
- Duration selected in semesters: 1 semester, 2 semesters, or 1 academic year.
- Proximity warning displayed if selected institution is far from the accommodation.
- No visit required. Offer, acceptance, and agreement process follows directly.
- Gender filtering applied silently throughout — female seekers only see female-tagged rooms.

6. Listing Page

The listing page is the primary conversion surface of the platform. Its design reference is Airbnb's listing detail view. The page is accessible to unauthenticated visitors with the exception of the BeeBop Valuation Report, which is gated behind login as a registration incentive.

6.1 Page Components

Component	Detail	Type Variation
Image carousel	Horizontally scrollable at the top of the page. Clicking any image opens the full gallery.	All types
Gallery	Full-screen photo gallery sectioned by room labels defined by the landlord (Living Room, Bedroom, Kitchen, etc.). Inspector-captured walkthrough footage appears as a separate BeeBop Verified Walkthrough section — clearly distinguished from landlord photos.	All types
Title	Mandatory property name.	All types
Subtitle	Optional secondary description.	All types
Star rating and review count	Displayed as e.g. 4.8 (34 reviews). Review count links to reviews carousel at page bottom.	All types except Sales
Verification badges	Document badge and Physical badge displayed inline. If both are present, replaced by consolidated Fully Verified badge.	All types
Award badges	Platform-awarded badges: Guest Favourite (short-let), Highly Responsive (all), Verified Landlord (all). Awarded automatically by the platform based on performance thresholds — never purchased or manually assigned.	Type-dependent
Host display	Host first name, avatar, time on BeeBop, response rate, and average response time.	Short-let only
Description	Free-text by lister. Minimum 200 characters. Collapses beyond 1,000 characters with a Read more expansion.	All types
Amenities	Structured checklist grouped by Power, Water, Security, Internet, Parking, Kitchen, Laundry. Amenities confirmed by inspector carry a Confirmed indicator.	All types
BeeBop Valuation Report	Gated behind login. Contains: area infrastructure scores, inspector professional note, and report date. Does not contain fair market value indicator or comparable listings.	All types — gated
Location map	Approximate pin (estate/street level — not exact gate address). Exact address revealed only after offer	All types

	acceptance or visit confirmation. Nearby landmarks drawn from area scoring data.	
Duration selector	Type-specific booking interface embedded in the fixed CTA bar.	Type-dependent
Award badge section	Dedicated section explaining each earned award badge.	All types
Customer review carousel	Verified tenants and guests only. Short-let cards include sub-category ratings: Accuracy, Cleanliness, Location, Value.	All types except Sales

6.2 Duration Selector — Type-Specific

Type	Duration Interface	CTA
Short-let	Date picker (check-in / check-out). Price updates dynamically. Minimum stay enforced.	Reserve
Student accommodation	Institution selector (pre-filtered to accepted institutions). Semester count: 1 semester, 2 semesters, 1 academic year.	Send offer
Rent	Duration in years (1 year, 2 years). Total cost displayed.	Send offer / Schedule a visit
Sales	No duration selector.	Send offer / Schedule a visit

6.3 BeeBop Valuation Report

The valuation report is the listing page's most distinctive feature. It is generated from inspector assessment data and area scoring records. It contains three sections only.

Report Section	Content
Area infrastructure scores	Road condition, NEPA/PHCN supply, security, and proximity scores with the date of last assessment clearly displayed.
Inspector's professional note	A short, neutral written observation from the physical assessment — factual and descriptive, not evaluative of price.
Report date	Always visible. Seekers can assess the currency of the assessment.

The report does not contain a fair market value indicator or comparable listings. It is a trust and transparency tool — BeeBop has been here, here is what they found — not a market analysis tool.

Unauthenticated visitors see a locked panel describing the report contents and a registration prompt. This is a deliberate registration incentive positioned at the moment the seeker is most invested in the listing.

6.4 Fixed CTA Bar

The call-to-action bar is fixed to the bottom of the viewport and remains visible regardless of scroll position. It contains the duration selector, the calculated price, and the action button. For unauthenticated visitors the button reads Sign in to continue — keeping registration as a visible and persistent prompt rather than a hidden barrier.

7. Short-Let In-Booking Messaging

Short-let bookings require a communication channel between seeker and host for questions about the property, additional services, or booking-specific arrangements. However, unrestricted direct communication creates a risk of seekers and hosts transacting off-platform to avoid fees.

BeeBop addresses this with a moderated group chat model. Every confirmed short-let booking automatically creates a group chat thread containing three participants: the seeker, the host, and a BeeBop moderator account. The moderator is the control mechanism — their presence creates accountability and the chat log is a complete, platform-held record of all communications.

7.1 Chat Architecture

Element	Detail
Thread creation	Automatic on booking confirmation. Three participants: seeker, host, BeeBop moderator. Thread linked to the specific booking record — scoped to this booking only.
Messaging	Real-time via WebSocket. Messages stored in database. Offline participants see messages on next connection.
Chat access	Accessible from booking card in seeker dashboard and host booking management panel. Opens inline — no separate page navigation required.
On-platform action shortcuts	When a seeker asks about extending a stay, a system-detected intent renders an Extend booking inline action button in the chat. Other shortcuts: Report issue, Request early check-in. Actions execute on-platform — no redirect to WhatsApp or external contact.
Scope restriction	In-booking group chat is strictly short-let only. Rent, sale, and student accommodation do not have chat threads.

7.2 Content Filtering and Moderation

Layer	Detail
Content filtering — primary detection	Python regex patterns monitor all messages for: Nigerian phone number formats (080x, 070x, 081x, 090x, 091x, 071x), 10-digit account number strings, Nigerian bank names, and common off-platform payment phrases. Flagged messages are delivered but marked internally and trigger a moderator alert.

Moderator alert queue	Flagged messages appear in the admin portal moderation queue within 30 seconds. Moderator sees full conversation context.
Moderator actions	Dismiss (false positive — clears flag). Warn participant (sends a BeeBop system warning message in the chat thread). Escalate (suspends participant account pending review and creates an admin action record).
Moderator presence	Moderator can read all messages across all chat threads, send messages as BeeBop, and take action at any time. Moderator messages displayed with the BeeBop label in the chat.

8. Dashboards

8.1 Seeker Dashboard

Accessible from the sidebar navigation. Three sections.

Section	Detail
Saved listings	Grid of bookmarked listings. Each card shows image, title, badge status, price, and a View action. Listings that become let agreed, sold, or delisted remain in the saved list but are visually marked as unavailable with a clear status label.
Active offers	List of offers currently in flight. Each row shows the property, offered price, current status (sent, counter-received, accepted, rejected), and submission date. Counter-offers awaiting response are highlighted with a visible 48-hour response timer.
Agreements	Signed agreements the seeker is a party to. Each shows property, agreement dates, days remaining in tenancy, and PDF download link. Renewal prompt appears 30 days before expiry where the landlord has indicated openness to renewal.

8.2 Landlord Dashboard

Section	Key Information
Overview panel	Active listing count, pending actions count (red badge — clears as actions are completed), unread notification count.
My listings	Listing cards with status: Draft, Under doc review, Live unverified, Doc verified, Fully verified, Let agreed / sold. Each state has distinct visual treatment.
Offers and enquiries	New offers received, counter-offers in progress, accepted and rejected offers, expired offers. Each offer card shows seeker verification status, offered vs listed price, and three actions: Accept, Counter, Reject. 48-hour response timer visible.
Visits	Read-only for landlords. Shows upcoming visits with date, time, and BeeBop agent name. Landlord can flag a scheduling conflict — BeeBop reschedules, not the landlord.
Agreements	Pending signatures, signed and active agreements, expiring within 60 days, and past agreements. Renewal fee prompt at correct threshold.
Listing analytics	Total views, saves, enquiry conversion rate, and average time from listing to offer per property.
Fees and billing	Itemised facilitation fees by transaction. No listing fees appear here as there are none. Payment history with dates and linked listings.

8.3 Student Accommodation Dashboard — Property Management System

The student accommodation dashboard operates at bed level, functioning as a lightweight property management system.

Unit Type Management

The inventory panel displays all unit types: single room, 2-in-a-room, 3-in-a-room, self-contain, and any custom types. Each shows total units, total beds, and a gender breakdown (Female x N, Male x N). Self-contain units show no gender tag — they are private units with no shared occupancy.

Room-Level Inventory Grid

Selecting a unit type opens a room-by-room grid. Each row shows room name, gender tag, and individual bed status indicators: Occupied, Available, or Vacating. Rooms with available beds display a View enquiries action. Gender tag can be edited only when the room is fully vacant — enforced at database level, not just UI.

Gender Tagging

Gender tags live at room level. Seekers are shown only rooms matching their registered gender. This filtering happens silently — the seeker never sees a male room even shown as unavailable. The waitlist displays gender breakdowns per unit type so the landlord can see which gender their available inventory needs to serve.

8.4 Short-Let Dashboard

Section	Detail
Availability calendar	14-day rolling calendar (scrollable to 90 days). Day states: Available, Booked, Turnover. Booked and Turnover days link to the relevant booking record.
Upcoming bookings	Booking cards showing check-in and check-out dates, duration, price, payment status, and days until check-in.
Booking requests	Pending requests with Accept and Decline actions. Accept captures payment; Decline notifies seeker.

Pricing settings	Base nightly rate, weekend rate, minimum stay, turnaround window duration, instant booking toggle.
Analytics	Bookings this month, revenue, occupancy rate, and average guest rating.

9. Search and Filtering

9.1 Two Discovery Modes

BeeBop offers two discovery modes for two distinct browsing mindsets. The conversational mode serves seekers who know what they want and can describe it. The category browse mode serves seekers who want to explore and filter manually. The two modes share state — a conversational query can seed a category browse, and a category page contains a prompt to continue with AI assistance.

9.2 Category Browse Structure

Each sidebar category page (Off-campus, Short-let, Rent, Sales) has a left filter panel, a listing grid, and a map toggle. Filters remain active on the map view. Pins are colour-coded by verification status: teal for fully verified, blue for doc verified, grey for unverified.

9.3 Filters — Shared Across All Categories

Filter	Detail
Location	Searchable by district, estate, or landmark. Multi-select.
Verification status	Checkboxes: Fully Verified, Doc Verified, Physically Inspected, Unverified. Default state pre-checks Fully Verified and Doc Verified — nudging seekers toward verified listings from first browse.
Amenities	Collapsible checklist: Power, Water, Security, Internet, Parking, Kitchen, Laundry.
Sort	Relevance (default), price low to high, price high to low, newest listing, highest rated.

9.4 Category-Specific Filters

Category	Filters
Off-campus	Institution (pre-applied from seeker profile). Gender (silently pre-applied from seeker profile — not shown as a user control). Unit type: single room, 2-in-a-room, 3-in-a-room, self-contain. Payment structure: per semester, per academic year. Available now toggle.

Short-let	Date range (check-in and check-out) — sits at top of filter panel. Number of guests. Minimum stay: any, 1 night, 3 nights, 1 week. Instant booking toggle. Minimum rating: 3+, 4+, 4.5+.
Rent	Bedroom count (multi-select). Property type: flat, detached, semi-detached, terraced, BQ, mini flat, self-contain. Furnishing: furnished, semi-furnished, unfurnished. Payment structure: annual or 2 years upfront. Available from date picker. Price range slider.
Sales	Bedroom count. Property type: detached, semi-detached, terraced, flat, land only, commercial. Development status: ready, off-plan, under construction. Title type: C of O, Governor's Consent, Deed of Assignment, Leasehold. Price range slider in 1m Naira increments.

10. AI and NLP Layer

The AI and NLP layer converts natural language property queries into structured database searches. It processes every query through a five-stage pipeline and maintains session-only context — there is no persistent memory across sessions.

10.1 Intent Classification

Intent	Example	Response
Search	I need a 2-bed in Wuse 2 under 300k	Extract parameters and query the database.
Clarification	How many rooms does that second one have?	Reference current session result set and answer contextually.
Information	Is Gwarinpa safe?	Answer from the area knowledge base. No search triggered.
Transactional	I want to schedule a visit to the third one	Route to the relevant platform action.

10.2 Nigerian Housing Vocabulary Map

The NLP layer maintains a curated vocabulary map included in the system prompt on every API call. Key mappings:

Term	Mapped Meaning
Self-con / self-contain	Self-contained unit with private bathroom and kitchen.
BQ / boys quarters	Small separate structure on a compound, typically one room.
Face-me-I-face-you	Shared corridor housing with multiple rooms opening to a common passage.
Mini flat	Open-plan studio with a separate bathroom.
2-in-1 / 3-in-1	Shared room with 2 or 3 beds.
Compound house	Property where multiple tenants share a compound.
Distress sale	Property sold urgently, often below market value.
400 (in rental context)	Interpreted as 400,000 Naira per year, not 400 Naira.
Behind / off / close to [landmark]	Location descriptor — mapped to GPS proximity search.

10.3 Parameter Extraction

After intent classification and vocabulary normalisation, the query is decomposed into structured parameters. Partial information is handled gracefully — a search with only location stated is valid. The layer extracts what is present, runs the search, and prompts for the single most useful missing parameter rather than demanding everything upfront.

10.4 Session Memory

The conversational interface is stateful within a single session. Session context — current search parameters, last result set (listing IDs), and last 10 conversation turns — is stored in Redis with a session-scoped key. The key expires after 30 minutes of inactivity or when the user clicks New Chat. There is no persistent chat history across sessions and no Chats sidebar section. Each new session starts completely fresh.

This session-only model enables reference resolution within a conversation — the user can say the third one or show me ones with a generator and BeeBop correctly understands the reference — without the storage cost or complexity of cross-session persistence.

10.5 Result Ranking

Signal	Weight	Notes
Verification status	Highest	Fully verified above doc verified above unverified. Non-negotiable — drives the verification behaviour BeeBop requires from landlords.
Rating score	Second	High average rating from verified tenants or guests. A listing with a 4.8 rating ranks above an unrated listing of the same verification tier.
Parameter match precision	Third	How closely the listing matches the extracted parameters from the query.
Recency	Fourth	Recently updated listings above stale ones.
Price competitiveness	Light	Listings priced at or below area average receive a small boost. Not displayed to the user — used only in ranking.
Saves	Not used	Removed. Noisy signal — saves reflect curiosity, not quality.

Enquiries	Not used	Removed. Can reflect misleading listings as easily as good ones.
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10.6 LLM Architecture

Model	Role	Rationale
Claude (Anthropic) — claude-sonnet-4-6	Primary engine. NLP query parsing, parameter extraction, conversation memory, valuation report generation.	Extended context window. Reliable structured JSON output. Strong performance on complex, culturally specific language tasks.
GPT-4o (OpenAI)	Fallback engine. Image analysis tasks.	Vendor resilience. Native multimodal capability for any image-based features.

A query log is maintained from day one: every query, extracted parameters, result IDs returned, and seeker click-through behaviour. This log is the foundation for ongoing vocabulary map expansion and NLP quality improvement.

11. Fee Structure

All listings are free to upload. BeeBop earns exclusively at the point of transaction completion. There are no upfront listing fees, document submission fees, or inspection fees charged to listers.

11.1 Rent

Tier	Annual Rent	Landlord Pays	Seeker Pays	Total BeeBop Revenue
Standard	Up to 5,000,000 Naira	15,000 Naira flat + 2% of annual rent	5,000 Naira flat + 1% of annual rent	20,000 Naira + 3% of annual rent
Premium	Above 5,000,000 Naira	15,000 Naira flat + approx. 1.34% of annual rent	5,000 Naira flat + approx. 0.66% of annual rent	20,000 Naira + 2% of annual rent

Rent Fee Notes

Flat fee always split: landlord pays 15,000 Naira, seeker pays 5,000 Naira, regardless of tier. Percentage components maintain approximately a 2:1 landlord-to-seeker ratio. No cap at any rent level. All fees collected at agreement signing via Paystack.

11.2 Student Accommodation

Party	Fee	Timing
Accommodation owner	2.5% of agreed accommodation price for the agreed term	Collected at agreement signing via Paystack
Seeker (student)	3% of agreed accommodation price for the agreed term	Collected at agreement signing via Paystack
Total BeeBop revenue	5.5% of agreed accommodation price	Both charges are independent — not pooled or split

11.3 Short-Let

Tier	Booking Value	Host Pays	Seeker Pays	Total BeeBop Revenue
Standard	Up to 150,000 Naira	3% of booking value	10% of booking value	13% of booking value
Premium	Above 150,000 Naira	3% of booking value	8% of booking value	11% of booking value

Short-Let Fee Notes

Booking value is the total stay amount (nightly rate × nights), not the per-night rate. Threshold check runs at payment capture time. Host payout is gross booking value minus 3% host fee, disbursed after check-in confirmation.

11.4 Sales

Tier	Sale Value	Seller Pays	Buyer Pays	Total BeeBop Revenue
Standard	Up to 50,000,000 Naira	3% of sale value	Zero	3% of sale value
Premium	Above 50,000,000 Naira	2.5% of sale value	Zero	2.5% of sale value

Sales Fee Notes

Buyers pay zero platform fees — a strong acquisition message. BeeBop generates a Paystack invoice to the seller at agreement signing. The signed agreement PDF is withheld until invoice payment is confirmed. Seller has 48 hours to pay before the case is escalated to admin review.

11.5 Renewal Fees

Type	Fee	Paid By	Trigger
Rent renewal	10,000 Naira flat	Landlord	On tenancy renewal. No new inspection required. Simplified agreement flow.
Student renewal	Per-semester or annual fee recalculated at current rate	Both parties	On renewal at end of agreed term.

Short-let	No renewal fee — new booking created for each stay	N/A	N/A
Sales	No renewal — one-time transaction	N/A	N/A

12. Notification System

12.1 Channels

Channel	Priority	Use Cases
Email (Resend)	Primary for OTP verification and formal documents	Verification OTPs, agreement PDFs, fee receipts, onboarding confirmations, monthly summaries.
WhatsApp Business API	Primary for time-sensitive transaction alerts	Offer received, offer accepted, visit confirmed, agreement ready to sign, badge issued. Uses approved Meta message templates.
In-app (WebSocket)	Real-time within platform	All events. Complements email and WhatsApp rather than replacing them for high-priority events.

SMS has been removed from the notification stack entirely. Email and WhatsApp together provide higher delivery reliability than SMS in Nigeria, lower cost, and remove the regulatory requirement for NCC DND registration.

12.2 Key Notification Triggers

Event	Channels	Notes
New offer received (landlord)	WhatsApp + in-app	48-hour response window starts at receipt.
Offer accepted	WhatsApp + email + in-app	Both parties notified. Agreement generation triggered.
Offer rejected	In-app only	Lower urgency. WhatsApp unnecessary.
Counter-offer received	WhatsApp + in-app	Response timer visible in notification.
Visit confirmed	WhatsApp + in-app	Date, time, and agent name included.
Agreement ready to sign	WhatsApp + in-app	High priority for both parties.
Agreement signed	Email (with PDF) + in-app	Formal document delivery via email.
Badge issued (landlord)	WhatsApp + in-app	Doc or physical badge awarded.

Short-let chat message flagged	In-app (moderator queue)	Moderator alerted within 30 seconds.
Tenancy expiring in 30 days	In-app + email	Renewal prompt if landlord has indicated openness.

12.3 Offer Response Timer Escalation

Time Since Event	Notification
Hour 0	WhatsApp + in-app. Offer or counter received.
Hour 24	In-app reminder only.
Hour 36	WhatsApp reminder: You have 12 hours remaining to respond.
Hour 48	WhatsApp + in-app to both parties confirming offer expiry.

13. Trusted Agent Workflow

Trusted agents are BeeBop's physical representatives in the real world. Their presence at a property visit is what enforces the no-direct-contact policy and keeps the transaction on-platform. Agents are onboarded by admin — there is no self-registration.

13.1 Assignment

Visit requests appear in the admin portal visit queue. Admin manually selects an available agent and assigns them. There is no automated assignment engine at this stage — manual assignment gives admin full control and allows judgment calls (area familiarity, listing type, agent experience). The agent is notified via WhatsApp and in-app and has two hours to confirm or flag a conflict.

13.2 Briefing Pack

Before every visit the agent receives a briefing pack in the internal portal: property address with GPS, listing photos for comparison, verification status, seeker first name and any noted preferences, and a conduct reminder checklist. Accessible on mobile.

13.3 Visit Conduct Standards

Standard	Requirement
Timing	Agent arrives five to ten minutes before the seeker.
Landlord presence	Landlord should be absent. Agreed during the scheduling call. Agent manages interaction if landlord insists on attending.
Walkthrough	Every room that appears in the listing photos is shown. Briefing pack photos used as checklist.
Observations	Discrepancies between listing and property noted silently for the post-visit report — not communicated to the seeker during the visit.
Questions	Answered from briefing pack. Questions that cannot be answered are noted and relayed through the platform post-visit.
Contact	Agent never shares personal contact details, landlord contact details, or seeker contact details with any party. Any attempt to exchange contacts is politely intercepted.

Closing	Agent does not linger. No informal conversations facilitated between seeker and any property-side party.
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13.4 Post-Visit Report

Within two hours of the visit, the agent submits a post-visit report covering: visit confirmation (did the visit take place, access issues, conduct issues) and property observations (amenity checklist vs listing, discrepancy notes, free-text observations). Report enters the admin review queue. On admin approval, the seeker's mandatory post-visit offer is unlocked.

13.5 Role Separation

A staff member may hold both Inspector and Trusted Agent roles on the platform. However, the same individual cannot be both the inspector and the visit agent for the same listing. This is enforced at the assignment level in the admin portal and at the API level — preventing any implied stake in the visit outcome.

14. Rating and Review System

Reviews on BeeBop are restricted to verified participants — seekers who completed a BeeBop-facilitated transaction on that specific listing. Open reviews that anyone can post are gamed within weeks and destroy the trust signal reviews are meant to provide.

14.1 Eligibility

Listing Type	Reviewer Eligibility	Review Prompt Timing
Short-let	Guests who completed a confirmed booking and check-in.	24 hours after check-out.
Student accommodation	Tenants who signed a BeeBop-facilitated agreement for a room in that property.	30 days after agreement signing.
Rent	Tenants who signed a BeeBop-facilitated tenancy agreement on that listing.	30 days after agreement signing.
Sales	Not applicable. No repeat-guest dynamic on a one-time transaction.	N/A

14.2 Short-Let Sub-Category Ratings

Short-let guests rate both an overall star score (1–5) and four sub-category scores: Accuracy, Cleanliness, Location, and Value. Sub-category averages are displayed on the listing page alongside the overall rating and feed into the award badge logic.

14.3 Award Badge Automation

Badge	Eligibility Criteria	Mechanism
Guest Favourite	Short-let listing with 4.5+ average rating from minimum 10 reviews in last 12 months.	Nightly automated job. Revoked automatically when threshold no longer met.
Highly Responsive	Landlord with 90%+ offer response rate within 24 hours over last 30 days.	Nightly automated job. Applies to all listing types.
Verified Landlord	NIN verification confirmed and profile photo uploaded.	Issued at onboarding completion. Displayed on lister profile.

15. The BeeBop Knowledge Base

Every physical inspection generates a structured data record contributing to BeeBop's growing knowledge base of Abuja's residential infrastructure. This is a strategic asset that compounds in value with every inspection conducted.

15.1 What is Captured

Data Type	Captured At	Compounding Value
Property-level assessment	Each inspection and annual re-inspection	A verified inventory of Abuja's residential properties with professional condition ratings.
Area infrastructure scores	Each inspection and any admin-initiated update	Tracks road condition, electricity supply, security, and proximity per geographic area over time.
Price history	Each listing event, price change, and completed transaction	A verified price database per area and property type.
Occupancy patterns	Each let-agreed and renewal event	Reveals demand patterns per area, season, and property type.
Search query data	Every AI query submitted	Shows how Abuja residents describe and prioritise housing — invaluable for vocabulary expansion.

15.2 Downstream Applications

- Annual State of Abuja Housing Infrastructure reports published publicly — establishing BeeBop as the authoritative research voice on residential property in the capital.
- Area intelligence pages on the platform showing neighbourhood trends and infrastructure change over time.
- Standalone property valuation product, backed by the surveying firm's assessments and the growing dataset.
- Institutional data products sold or licensed to urban planners, FCT authorities, development organisations, or commercial researchers.

16. Open Items and Decisions Pending

The following items have been identified during product design as requiring a final decision before the relevant features are built.

Item	Options	Recommendation
Rent premium tier — percentage split above 5m	Maintain 2:1 ratio (1.34% landlord, 0.66% seeker) or simplify (landlord pays full 2%, seeker pays flat 5,000 Naira only).	Simpler structure (landlord pays all percentage, seeker pays flat only) is easier to communicate and collect. Decide before Phase 2 Sprint 10.
Short-let booking cancellation policy	Host-initiated cancellation penalties. Seeker-initiated refund policy.	Define before Phase 2 Sprint 11. Common model: full refund if cancelled 7+ days before check-in, 50% within 7 days, no refund within 48 hours.
Sales agreement process	BeeBop-generated standard agreement vs. referring parties to independent solicitors.	BeeBop-generated is recommended for platform control. Requires lawyer sign-off on template before Phase 2 Sprint 10.
Unverified listing visibility	Show unverified listings in browse with lower ranking, or hold all listings until doc-verified.	Hold until doc-verified is recommended to protect platform trust from day one.
Inspector and agent staffing model	Employed staff vs. vetted contractors for both roles.	Legal and tax implications to confirm with counsel.
Short-let rating sub-categories for rent	Extend sub-category ratings (Accuracy, Cleanliness, etc.) to rent listings, not just short-let.	Adds data richness. Decision before Phase 4 rating sprint.

17. Glossary

Term	Definition
Area score	A set of infrastructure ratings (road condition, NEPA supply, security, proximity) associated with a geographic area, stored against GPS coordinates and updated independently of badge status.
Award badge	A platform-issued recognition badge awarded automatically based on performance thresholds. Never manually assigned and never purchasable.
BeeBop Valuation Report	An AI-generated assessment produced from inspector data and area scores. Contains area infrastructure scores, inspector professional note, and report date only. Does not contain fair market value comparisons.
BQ / Boys quarters	A small separate structure on a compound. Typically one room with a bathroom. Common secondary dwelling on Nigerian residential properties.
Content filtering	Automated regex pattern matching applied to short-let group chat messages to detect off-platform transaction attempts (phone numbers, account numbers, bank names).
Doc badge	The document verification badge, confirming that a listing's title documents have been reviewed and found credible by BeeBop admin.
Facilitation fee	The fee charged at agreement signing. Structure varies by listing category.
Face-me-I-face-you	Shared corridor housing common in Nigerian cities. Multiple rooms open onto a common passage. Distinct from self-contained or BQ units.
Fully Verified	Consolidated designation displayed when a listing has earned both the document badge and the physical badge.
Inspector	A staff member or contractor affiliated with BeeBop's surveying firm who conducts physical property assessments and files area infrastructure scores.
NIN	National Identification Number. 11-digit identifier issued by the National Identity Management Commission (NIMC). Used for landlord and agent identity verification.
Physical badge	The physical inspection badge, confirming that a BeeBop inspector has visited the property and found it to match its listing.
PWA	Progressive Web Application. Browser-based application with offline capability and background sync via service workers. Used for the inspector portal.
Self-con / self-contain	A self-contained residential unit with its own private bathroom and kitchen. Distinct from shared facilities.
Session context	The Redis-stored object holding the current conversation turns, search parameters, and result set for the active user session. Expires after 30 minutes of inactivity.
Trusted agent	A BeeBop staff member or vetted contractor who attends property visits as the platform's representative, ensuring no direct contact between seekers and landlords.

Turnaround window	A configurable blocked period between short-let bookings set by the host for cleaning and preparation.
WhatsApp Business API	Meta's programmatic API for sending WhatsApp messages using pre-approved templates. Used for time-sensitive notifications and OTP verification.